

PROBABILITY vs STATISTICS — CONCEPTUAL MCQs

1. You know that a fair die has six equally likely outcomes. You want to calculate the probability of getting a 6. This is an example of:

- A. Statistics
- B. Probability**
- C. Regression
- D. Sampling

2. You roll a die 1,000 times and record the results. You notice that the number 6 appeared much more frequently than expected. You want to determine whether the die may be biased. This is primarily:

- A. Probability
- B. Statistics**
- C. Linear Algebra
- D. Calculus

3. Which statement best describes Probability?

- A. Using observed data to estimate unknown characteristics
- B. Using a known model or assumptions to reason about possible outcomes**
- C. Organizing data into tables
- D. Calculating averages from collected data

4. Which statement best describes Statistics?

- A. Predicting possible outcomes from a known probability model
- B. Using observed data to understand or estimate an underlying process**
- C. Generating random numbers
- D. Calculating only probabilities between 0 and 1

5. A company knows that historically 2% of its products are defective. It wants to estimate how many defective products might appear in the next 10,000 products.

This is primarily:

- A. Statistics
- B. Probability**
- C. Data cleaning
- D. Visualization

6. A company collects data from 10,000 products and finds that 250 are defective. It wants to estimate the defect rate of the production process.

This is primarily:

- A. Probability
- B. Statistics**
- C. Linear Algebra
- D. Optimization

7. Which direction best represents Probability?

- A. Data → Model
- B. Model → Possible outcomes**
- C. Data → Visualization
- D. Sample → Database

8. Which direction best represents Statistics?

- A. Model → Possible outcomes
- B. Possible outcomes → Model
- C. Observed data → Understanding/inference about the underlying process**
- D. Probability → Randomness

9. A medical researcher knows that a disease occurs in approximately 1% of the population. She wants to calculate the probability that a randomly selected person has the disease.

This is primarily:

- A. Statistics
- B. Probability**
- C. Data preprocessing
- D. Regression

10. A medical researcher observes that 15 out of 1,000 randomly selected people have a particular disease. She wants to estimate the disease prevalence in the larger population.

This is primarily:

- A. Probability
- B. Statistics**
- C. Matrix algebra
- D. Optimization

11. Consider the following situation:

"A coin is believed to be fair. What is the probability of getting 7 Heads when the coin is tossed 10 times?"

This is:

- A. Statistics
- B. Probability**
- C. Descriptive statistics
- D. Sampling

12. Now consider:

"A coin is tossed 10 times and produces 9 Heads. Based on this observation, determine whether the coin is likely to be fair."

This is:

- A. Probability
- B. Statistics**
- C. Matrix multiplication
- D. Data visualization

13. Which question is a Probability question?

- A. What is the average height of students in this class?
- B. What is the probability that a randomly selected student is taller than 180 cm?
- C. What is the median height of the students?
- D. What is the standard deviation of student heights?

14. Which question is a Statistics question?

- A. What is the probability of getting Heads on a fair coin?
- B. What is the probability of rolling a 6?
- C. What is the average income observed in this dataset?
- D. What is the probability of rain tomorrow?

15. A weather model says that there is a 70% probability of rain tomorrow.

This statement is an example of:

- A. Statistics
- B. Probability
- C. Descriptive statistics
- D. Sampling

16. You collect the rainfall measurements from the last 20 years and calculate the average annual rainfall.

This activity is primarily:

- A. Probability
- B. Statistics
- C. Probability distribution
- D. Random sampling

Just to make it clearer – Probability and statistics

17. "What is likely to happen?"

- A. Probability
- B. Statistics

18. "What happened in the data?"

- A. Probability
- B. Statistics

19. "What can we learn about the population from our sample?"

- A. Probability
- B. Statistics

20. "If the assumptions are known, what outcomes should we expect?"

A. Probability

B. Statistics